

Continuous time Quantum Monte Carlo Method Note

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I. BACKGROUND INFORMATION

A. Vector space

In quantum mechanics, the complete physical information about a system is encoded in its *state*, represented **abstractly** by a ket

$$|\psi\rangle. \quad (1)$$

I really want to highlight the word abstractly here, because the quantum state acts as an exhaustive "container" for all the potential information that can be known about a particle or system.

Discrete basis $\{ u_i\rangle\}$	Continuous basis $\{ w_{\mathbf{q}}\rangle\}$
$\langle u_i u_j \rangle = \delta_{ij}$	$\langle w_{\mathbf{q}} w_{\mathbf{q}'} \rangle = \delta(\mathbf{q} - \mathbf{q}')$
$\sum_i u_i\rangle \langle u_i = \mathbb{1}$	$\int d\mathbf{q} w_{\mathbf{q}}\rangle \langle w_{\mathbf{q}} = \mathbb{1}$
$ \psi\rangle = \sum_i c_i u_i\rangle$	$ \psi\rangle = \int d\mathbf{q} \psi(\mathbf{q}) w_{\mathbf{q}}\rangle$
$c_i = \langle u_i \psi \rangle$	$\psi(\mathbf{q}) = \langle w_{\mathbf{q}} \psi \rangle$
$\langle \phi \psi \rangle = \sum_i b_i^* c_i$	$\langle \phi \psi \rangle = \int d\mathbf{q} \phi^*(\mathbf{q}) \psi(\mathbf{q})$
$\langle \psi \psi \rangle = \sum_i c_i ^2$	$\langle \phi \psi \rangle = \int d\mathbf{q} \phi(\mathbf{q}) ^2$

Example 1: The two-state quantum system has two independent states (dimension 2), we can denote them by $|E_0\rangle$ and $|E_1\rangle$.

Example 2: Position space $\{|\mathbf{r}\rangle\}$ is a continuous basis (infinite dimension).

These quantum states can be represented differently depending on the choice of basis (just like a 3D vector appears differently on Cartesian vs polar vs spherical coordinates).

Returning to the two state system, one convenient representation is using two column vectors $|E_0\rangle \rightarrow (1, 0)^T$ and $|E_1\rangle \rightarrow (0, 1)^T$.

An arbitrary quantum state of the system at time t is $|\psi(t)\rangle$, so in the position basis

$$|\psi(t)\rangle = \int d\mathbf{r} \psi(\mathbf{r}, t) |\mathbf{r}\rangle \quad (2)$$

Hence, what we usually call wavefunction that appears in the Schrodinger equation is

$$\psi(\mathbf{r}, t) = \langle \mathbf{r} | \psi(t) \rangle, \quad (3)$$

which is just the coefficient in the expansion over a particular set of basis vectors; not the state itself. It is also possible to from one orthonormal basis to another orthonormal basis. For example the momentum basis $\{|\mathbf{p}\rangle\}$. Position states $|\mathbf{r}\rangle$ and momentum states $|\mathbf{p}\rangle$ are two complete bases describing the same Hilbert space. we know that

$$\begin{aligned} \langle \mathbf{r} | \mathbf{p} \rangle &= \frac{1}{\sqrt{2\pi\hbar}} e^{i\mathbf{p}\cdot\mathbf{r}/\hbar} \\ \langle \mathbf{p} | \mathbf{r} \rangle &= (\langle \mathbf{r} | \mathbf{p} \rangle)^* = \frac{1}{\sqrt{2\pi\hbar}} e^{-i\mathbf{p}\cdot\mathbf{r}/\hbar} \end{aligned} \quad (4)$$

$$\begin{aligned} |\psi(t)\rangle &= \int d\mathbf{r} \psi(\mathbf{r}, t) \mathbb{1} |\mathbf{r}\rangle = \int d\mathbf{r} \int d\mathbf{p} \psi(\mathbf{r}, t) |\mathbf{p}\rangle \langle \mathbf{p} | \mathbf{r} \rangle \\ &= \int d\mathbf{p} \underbrace{\left(\frac{1}{\sqrt{2\pi\hbar}} \int d\mathbf{r} \psi(\mathbf{r}, t) e^{-i\mathbf{p}\cdot\mathbf{r}/\hbar} \right)}_{\text{Fourier transform}} |\mathbf{p}\rangle \\ &= \int d\mathbf{p} \tilde{\psi}(\mathbf{p}, t) |\mathbf{p}\rangle. \end{aligned} \quad (5)$$

B. Introduction to Operators in Quantum Mechanics

Unlike classical mechanics, physical quantities are not assigned definite numerical values in general. Instead, they are extracted from the state through the action of *operators*. An *operator* is a mathematical object that acts on a state and produces another state:

$$\hat{A} : |\psi\rangle \longrightarrow \hat{A}|\psi\rangle. \quad (6)$$

Operators are linear maps on the space of states (a Hilbert space). A central postulate of quantum mechanics is:

"Every measurable physical quantity (observable) is represented by a Hermitian operator."

Examples include:

- Position: $\hat{X}$
- Momentum: $\hat{P}$

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- Energy (Hamiltonian): $\hat{H} = \hat{T} + \hat{V} = \frac{\hat{P}^2}{2m} + \hat{V}$

For example, a measurement of the observable $\hat{X}$ performed on state $|x\rangle$ will yield the value x with certainty. Here x is just a scalar measurable quantity. For a general state $|\psi\rangle$, the average outcome of many measurements of $\hat{X}$ is given by the expectation value

$$\begin{aligned}\langle\hat{X}\rangle &\equiv \langle\psi|\hat{X}|\psi\rangle = \int dx dx' \langle\psi|x\rangle\langle x|\hat{X}|x'\rangle\langle x'|\psi\rangle \\ &= \int dx dx' \psi^*(x)\langle x|x'|x'\rangle\langle x'|\psi\rangle \\ &= \int dx dx' \psi^*(x)x'\langle x|x'\rangle\psi(x') \\ &= \int dx dx' \psi^*(x)x'\delta(x-x')\psi(x') \\ &= \int dx \psi^*(x)x\psi(x) \\ &= \int dx x|\psi(x)|^2.\end{aligned}\quad (7)$$

$$\begin{aligned}\langle x|\hat{P}|\psi\rangle &= \int dp \langle x|p\rangle\langle p|\hat{P}|\psi\rangle \\ &= \frac{1}{\sqrt{2\pi\hbar}} \int dp e^{ipx/\hbar} \langle p|\psi\rangle \\ &= \frac{1}{\sqrt{2\pi\hbar}} \int dp p e^{ipx/\hbar} \langle p|\psi\rangle \\ &= \frac{1}{\sqrt{2\pi\hbar}} \int dp \left(-i\hbar \frac{d}{dx}\right) e^{ipx/\hbar} \tilde{\psi}(p) \\ &= \left(-i\hbar \frac{d}{dx}\right) \underbrace{\left[\frac{1}{\sqrt{2\pi\hbar}} \int dp e^{ipx/\hbar} \tilde{\psi}(p)\right]}_{\text{Inverse Fourier transform}} \\ &= \left(-i\hbar \frac{d}{dx}\right) \psi(x)\end{aligned}\quad (8)$$

Importantly, the derivative $-i\hbar(d/dx)$ form is not the operator itself, but it is the representation of the momentum operator in position space.

We can also have another abstract way to present an operator using its eigenstates (spectral decomposition). For example, in D -dimensional space,

$$\hat{P} = \int d^D \mathbf{p} p |\mathbf{p}\rangle\langle\mathbf{p}| \quad (9)$$

Now considering the Schrodinger equation in the operator form

$$\hat{H}|\psi\rangle = E|\psi\rangle, \quad (10)$$

meaning the Hamiltonian operator reads out the energy of the quantum state. In position basis, by contracting both sides with the bra $\langle x|$ we have

$$\langle x|\hat{H}|\psi\rangle = \left[-\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} + V(x)\right] \psi(x) = E\psi(x) \quad (11)$$

II. QUANTUM HARMONIC OSCILLATOR AND CREATION/ANNIHILATION OPERATORS

$$\hat{H}_{\text{SHO}} = \frac{\hat{P}^2}{2m} + \frac{1}{2}m\omega^2\hat{X}^2 \quad (12)$$

It is possible to solve the Schrodinger equation explicitly for this potential to obtain energy spectrum and wave function (see any modern physics textbook)

$$E_n = \left(n + \frac{1}{2}\right) \hbar\omega \quad (13)$$

where $n = 0, 1, \dots, \infty$. I.e.,

$$\hat{H}_{\text{SHO}}|n\rangle = E_n|n\rangle \quad (14)$$

Define

$$\hat{a} = \sqrt{\frac{m\omega}{2\hbar}} \left(\hat{X} + \frac{i}{m\omega}\hat{P}\right) \quad (15)$$

$$\hat{a}^\dagger = \sqrt{\frac{m\omega}{2\hbar}} \left(\hat{X} - \frac{i}{m\omega}\hat{P}\right) \quad (16)$$

They satisfies the following commutation relation (recall the canonical commutation, $[\hat{X}, \hat{P} = i\hbar]$)

$$[\hat{a}, \hat{a}^\dagger] = 1; \quad [\hat{a}^\dagger, \hat{a}, \hat{a}^\dagger] = \hat{a}^\dagger; \quad [\hat{a}^\dagger, \hat{a}, \hat{a}] = -\hat{a} \quad (17)$$

It is possible to show that the hamiltonian will become

$$\hat{H}_{\text{SHO}} = \left(\hat{a}^\dagger \hat{a} + \frac{1}{2}\right) \hbar\omega \quad (18)$$

and

$$[\hat{H}, \hat{a}^\dagger] = \hbar\omega \hat{a}^\dagger \quad (19)$$

$$[\hat{H}, \hat{a}] = -\hbar\omega \hat{a} \quad (20)$$

Therefore,

$$\hat{H}_{\text{SHO}}|n\rangle = \left(n + \frac{1}{2}\right) \hbar\omega|n\rangle \quad (21)$$

so what $\hat{a}^\dagger|n\rangle$ and $\hat{a}|n\rangle$ are?

$$\begin{aligned}\hat{H}_{\text{SHO}}(\hat{a}^\dagger|n\rangle) &= \left(\hat{a}^\dagger \hat{a} + \frac{1}{2}\right) \hbar\omega (\hat{a}^\dagger|n\rangle) \\ &= \left(\hat{a}^\dagger + \hat{a}^\dagger \hat{a}^\dagger \hat{a} + \frac{1}{2}\hat{a}^\dagger\right) \hbar\omega (|n\rangle) \\ &= \left(n + 1 + \frac{1}{2}\right) \hbar\omega (\hat{a}^\dagger|n\rangle)\end{aligned}\quad (22)$$

Evidently, $(n + 1 + \frac{1}{2}) \hbar\omega$ is the eigen energy of the $|n+1\rangle$ state, so $\hat{a}^\dagger|n\rangle$ is also an eigenstate of $\hat{H}_{\text{SHO}}$

$$\hat{a}^\dagger|n\rangle = \sqrt{n+1}|n+1\rangle \quad (23)$$

$\sqrt{n+1}$ is a normalization coefficient. Similarly,

$$\hat{a}|n\rangle = \sqrt{n}|n-1\rangle \quad (24)$$

You can raise the state to infinity but cannot lower the state forever, the vacuum state $|0\rangle$

$$a|0\rangle = 0 \implies |n\rangle = \frac{[\hat{a}^\dagger]^n}{\sqrt{n!}}|0\rangle \quad (25)$$

Any energy eigenstate state of a single harmonic oscillator can be obtained from $|0\rangle$ by successively applying the creation operator $\hat{a}^\dagger$. Any state of the system can be expressed as a linear combination of the states. In addition, every operators can be expressed in terms of $\hat{a}$ and $\hat{a}^\dagger$. For example, the time evolution

$$\begin{aligned} \hat{U}(t) &= \exp(-it\hat{H}/\hbar) = \exp[-it\omega(\hat{a}^\dagger\hat{a} + 1/2)] \\ &= e^{-it\omega/2} \sum_{k=0}^{\infty} \frac{(e^{-i\omega t} - 1)^k}{k!} [\hat{a}^\dagger]^k [\hat{a}]^k \end{aligned} \quad (26)$$

A. Extending this ideal to general quantum theory

Introducing the creation operator $\hat{a}_{\mathbf{p}}^\dagger$ of a particle with a momentum $\mathbf{p}$. If $|0\rangle$ is the vacuum state

$$|\vec{\mathbf{p}}\rangle = \hat{a}_{\mathbf{p}}^\dagger|0\rangle \quad (27)$$

$$[a_{\vec{\mathbf{p}}}, a_{\vec{\mathbf{q}}}^\dagger] = (2\pi)^3 \delta^{(3)}(\vec{\mathbf{p}} - \vec{\mathbf{q}}) \quad (28)$$

Momentum operator

$$\hat{\mathbf{P}} = \int \frac{d^3\vec{\mathbf{p}}}{(2\pi)^3} \vec{\mathbf{p}} a_{\vec{\mathbf{p}}}^\dagger a_{\vec{\mathbf{p}}} \quad (29)$$

Indeed

$$\begin{aligned} \hat{\mathbf{P}}|\vec{\mathbf{p}}\rangle &= \int \frac{d^3\vec{\mathbf{q}}}{(2\pi)^3} \vec{\mathbf{q}} a_{\vec{\mathbf{q}}}^\dagger a_{\vec{\mathbf{q}}} a_{\vec{\mathbf{p}}}^\dagger|0\rangle \\ &= \int \frac{d^3\vec{\mathbf{q}}}{(2\pi)^3} \vec{\mathbf{q}} a_{\vec{\mathbf{q}}}^\dagger (2\pi)^2 \delta^{(3)}(\vec{\mathbf{q}} - \vec{\mathbf{p}})|0\rangle \\ &= \vec{\mathbf{p}} a_{\vec{\mathbf{p}}}^\dagger|0\rangle \\ &= \vec{\mathbf{p}}|\vec{\mathbf{p}}\rangle \end{aligned} \quad (30)$$

Free Hamiltonian in momentum space

$$\hat{H} = \int \frac{d^3\vec{\mathbf{p}}}{(2\pi)^3} E_{\vec{\mathbf{p}}} a_{\vec{\mathbf{p}}}^\dagger a_{\vec{\mathbf{p}}} \quad (31)$$

Free Hamiltonian in position space

$$\hat{H} = \int d^3\vec{\mathbf{x}} c^\dagger(\vec{\mathbf{x}}) \left(-\frac{\hbar^2}{2m} \frac{d}{dx} \right) c(\vec{\mathbf{x}}) \quad (32)$$

In general, you can write any single-particle operator as

$$\begin{aligned} \hat{A} &= \int d\alpha d\beta |\alpha\rangle\langle\alpha| \hat{A} |\beta\rangle\langle\beta| \\ &= \int d\alpha d\beta \mathcal{A}_{\alpha\beta} |\alpha\rangle\langle\beta| \equiv \int d\alpha d\beta \mathcal{A}_{\alpha\beta} a_{\alpha}^\dagger a_{\beta} \end{aligned} \quad (33)$$

III. INTRODUCTION

The most obvious way to construct a Monte Carlo (MC) simulation scheme for a lattice Hamiltonian H is to start from the path integral formulation of quantum statistical mechanics, where $e^{-\beta H}$ is written as a product of imaginary-time evolution operators $e^{-\varepsilon H}$, with a small “time slice”, $\varepsilon = \beta/L_\tau$ for some large number L_τ of slices [1, 2]

$$T(\beta) = e^{-\beta H} = \underbrace{e^{-\varepsilon H} \cdots e^{-\varepsilon H}}_{L_\tau \text{ terms}} \quad (34)$$

In this algorithm, it is possible to take the limit $\varepsilon \rightarrow 0$ and $L_\tau \rightarrow \infty$.

Then the powers of the Hamiltonian can be further decomposed into strings of two-body (bond) operators, $H_{b_n} \cdots H_{b_2} H_{b_1}$ with all combinations of the operators $H_b = H_{i,j}^{(b)}$, $b \in \{1, \dots, N_b\}$ on a finite lattice in any number of spatial dimensions D (with periodic boundary condition). The total number of bond a D -dimensional hypercubic lattice with L sites is $N_b = DL^D$. In this limit, we can expand

$$e^{-\varepsilon H} \simeq (1 - \varepsilon H) + \mathcal{O}(\varepsilon^2) \quad (35)$$

From this formula, it looks like there is still a systematic error associated with time discretizations. However, we will show that in practice ε is essentially 0, hence continuous time,

The Hamiltonian reads

$$H = -t \sum_{\langle i,j \rangle} (c_i^\dagger c_j + c_j^\dagger c_i) \quad (36)$$

Next we can express the Hamiltonian as

$$H = \sum_b H_b. \quad (37)$$

where b stand for the bond, which is any $\langle i, j \rangle$ pair.

Possible bond operators are given in Fig. 1 Then,

$$e^{-\varepsilon H} \simeq \left(1 - \varepsilon \sum_b H_b \right) + \mathcal{O}(\varepsilon^2) \quad (38)$$

Substituting this into the Trotterized expression gives

$$\begin{aligned} Z_{fi} &= \langle f | T(\beta) | i \rangle \\ &= \sum_{[b]} \langle f | (-\varepsilon H_{b_{N_{\text{op}}}}) \cdots (-\varepsilon H_{b_2}) (-\varepsilon H_{b_1}) | i \rangle, \end{aligned} \quad (39)$$

where $[b]$ denotes a continuous-time configuration consisting of an ordered sequence of N_{op} bond operators inserted at imaginary times $0 < \tau_1 < \tau_2 < \cdots < \tau_{N_{\text{op}}} < \beta$.

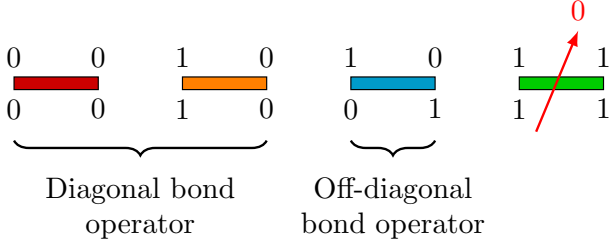


FIG. 1. Possible bond operators, where the 0 and 1 represent the fermion occupation at a particular site. There are two sets (above and below the bond operator) which indicate the state of the link before and after the bond operator is activated (added). In the free theory, the last operator has zero weight due to Pauli exclusion principle.

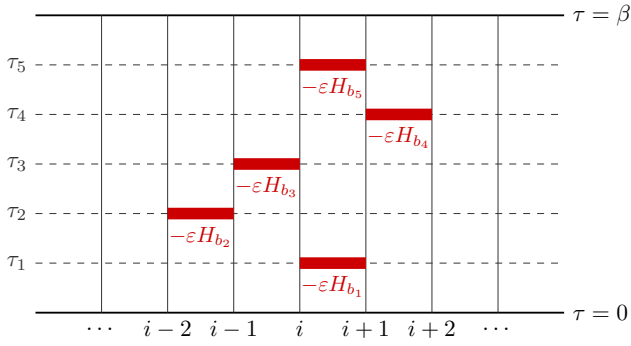


FIG. 2. Illustration of a continuous-time path-integral configuration $[b]$ as given in Eq. (39). The configuration shown contains $N_{\text{op}} = 5$ bond operators $(-\epsilon H_{b_i})$ at times τ_i . The bond without bond operator is 1. Notice that the number of particles is 0.

The path integral is formulated in a complete basis of the local Hilbert space, defined by the expression

$$\mathbb{1} = \sum_s |s\rangle\langle s| \quad (40)$$

where $\{|s\rangle\}$'s represent any discrete set of quantum numbers. Then

$$\begin{aligned} Z_{fi} &= \langle f|T(\beta)|i\rangle \\ &= \sum_{[b,s]} \langle f|(-\epsilon H_{b_{N_{\text{op}}}})|s_{N_{\text{op}}-1}\rangle \dots \langle s_2|(-\epsilon H_{b_2})|s_1\rangle \\ &\quad \times \langle s_1|(-\epsilon H_{b_1})|i\rangle, \\ &= \sum_{[b,s]} W([b,s]) \end{aligned} \quad (41)$$

Notice each quantity $\langle s_t|(-\epsilon H_{b_t})|s_{t-1}\rangle$ is a local matrix elements. $W[b, s]$ is the combined weight of the given configuration.

Let us consider the free Hamiltonian

$$H_0 = -t \sum_{\langle i,j \rangle} (c_i^\dagger c_j + c_j^\dagger c_i). \quad (42)$$

$$\langle 0, 0 | -\epsilon H_b | 0, 0 \rangle = 0 \quad (43)$$

$$\langle 1, 0 | -\epsilon H_b | 0, 1 \rangle = \epsilon t \quad (44)$$

$$\langle 0, 1 | -\epsilon H_b | 1, 0 \rangle = \epsilon t \quad (45)$$

$$\langle 1, 0 | -\epsilon H_b | 1, 0 \rangle = 0 \quad (46)$$

$$\langle 0, 1 | -\epsilon H_b | 0, 1 \rangle = 0 \quad (47)$$

$$\langle 1, 1 | -\epsilon H_b | 1, 1 \rangle = 0 \quad (48)$$

To make the quantum Monte Carlo (QMC) more efficient, we can add constant terms to the Hamiltonian. It is mathematically straightforward and does not alter the underlying physics, but it has important algorithmic implications. If the Hamiltonian is shifted as $H' = H + C$, then all eigenvalues shift uniformly according to $E'_n = E_n + C$, while the eigenstates remain unchanged. $e^{-\tau(H+C)} = e^{-\tau C} e^{-\tau H}$, introducing a global normalization factor. Therefore, the Hamiltonian that we actually study is

$$H_0 = -t \sum_{\langle i,j \rangle} (c_i^\dagger c_j + c_j^\dagger c_i - c_i^\dagger c_i - c_j^\dagger c_j + 2 \times \mathbb{1}). \quad (49)$$

$$\langle 1, 0 | -\epsilon H_b | 1, 0 \rangle = \epsilon t \quad (50)$$

$$\langle 0, 1 | -\epsilon H_b | 1, 0 \rangle = \epsilon t \quad (51)$$

$$\langle 1, 0 | -\epsilon H_b | 1, 0 \rangle = \epsilon t \quad (52)$$

$$\langle 0, 1 | -\epsilon H_b | 0, 1 \rangle = \epsilon t \quad (53)$$

$$\langle 1, 1 | -\epsilon H_b | 1, 1 \rangle = 0 \quad (54)$$

IV. THE MC ALGORITHM

A. Bond update

The total number of active bond operators is

$$N_{\text{op}} = n_{\text{diag}} + n_{\text{oddiag}} \quad (55)$$

We will try to insert/active a diagonal bond to the space-time lattice or remove an existing diagonal bond from the lattice. Each of these actions are equally likely. Therefore, half the time we will try to add and half the time we will try to remove

Step 1: Pick a random time slice (probability $1/L_\tau$) and a random bond (with probability $1/N_b$). **Personal choice: at the moment I only allow one bond operator existing at a particular time slice, so if there is already an operator, the code will exit and repeat.** Else, check the site and the neighbor's occupation number to determine type of diagonal bond can be added.

Step 2-a: Propose to add a diagonal operator with accept/reject probability P_{add} . The number of diagonal bond operators increases by 1 while the that of off-diagonal bond operators remain the same.

Step 2-b: If decide to remove a diagonal bond, pick one of them at random (probability $1/n_{diag}$). Propose to remove with accept/reject probability P_{rem}

How many times bond update are needed? Ideally, in each MC sweep each bond must be examined at least once, therefore N_{bond} times. This is the fixed number but when there are many active bonds operators, this can be inefficient.

1. Detailed balance for bond operator update

Let C be a specific worldline configuration of the lattice, and let C' be that exact same configuration but with one additional diagonal operator inserted at a specific bond b and time slice τ . To ensure the Monte Carlo simulation correctly samples the equilibrium distribution, the transition rates between state C (currently has n_{diag} diagonal operator) and state C' (more operators) must satisfy detailed balance.

The ratio of their statistical weights is simply the matrix element of the operator you are inserting.

$$\frac{W(C')}{W(C)} = \langle n'_i, n'_j | -\varepsilon H_b | n_i, n_j \rangle = \varepsilon w(n'_i, n'_j; n_i, n_j) \quad (56)$$

$$\begin{aligned} W(C)P_{prop}(C \rightarrow C')P_{acc}(C' \rightarrow C) \\ = W(C')P_{prop}(C' \rightarrow C)P_{acc}(C \rightarrow C'). \end{aligned} \quad (57)$$

Where

- P_{prop} is the a priori probability of proposing the update.
- P_{acc} is the probability of accepting the proposed update.

$$P_{acc}(C \rightarrow C') \equiv P_{add} \text{ and } P_{acc}(C' \rightarrow C) \equiv P_{rem}$$

$$P_{prop}(C \rightarrow C') = \frac{1}{2} \frac{1}{L_\tau} \frac{1}{N_b} \quad (58)$$

$$P_{prop}(C' \rightarrow C) = \frac{1}{2} \frac{1}{n_{diag} + 1} \quad (59)$$

Thus,

$$P_{add} = \frac{\varepsilon \overrightarrow{L_\tau}^\beta N_b w(n'_i, n'_j; n_i, n_j)}{n_{diag} + 1} P_{rem} \quad (60)$$

Indeed, the probability does not depend on time discretizations. When proposing to add an operator, we can set $P_{rem} = 1$. If $P < \min(1, P_{add})$, the proposal is accepted, else ejected.

Similarly, following the same procedure, we can show that

$$P_{rem} = \frac{n_{diag}}{\varepsilon \overrightarrow{L_\tau}^\beta N_b w(n'_i, n'_j; n_i, n_j)} P_{add} \quad (61)$$

When proposing to remove an existing operator, we can set $P_{add} = 1$. If $P < \min(1, P_{rem})$, the proposal is accepted, else ejected.

In this algorithm, only diagonal bond operator can be either added or removed, we will not modify the off-diagonal operator

B. Worm update

This algorithm will try to turn a diagonal bond operator into an off-diagonal one.

You can think the worm as particle carrier and can deposit the particle at a particular site and time slice.

This algorithm also has two possible action, either adding a particle or erasing a particle. Because the particle number of the system is conserved, erasing and adding need to complement each other.

The routine starts with throwing a coil and decide half the time you can try to add first and half the time you will erase first

In the add first:

1. Randomly select an empty (unoccupied) site at $\tau = 0$ time slice. Place the tail of the worm at that site, the particle number is flipped 0 to 1. The worm head will move forward in time. There are a few possibilities
 - (a) The worm head can move directly to $\tau = \beta$ without encountering any bond operator (See the first blue worldline from the left Fig. 6). After this step, the particle is temporary increased by one. Then randomly select an existing particle at $\tau = \beta$ and erase that particle (the worm head moves backward in time) until reaching $\tau = 0$. The move is completed. The particle number is restored and a new configuration is sampled.
 - (b) the worm head can move forward and encounter a bond operator at a time slice $0 < \tau < \beta$.
 - i. Fig. 3,
 - ii. Fig. 4 a,
 - iii. Fig. 5 a

Until the worm reach $\tau = \beta$. Then randomly select an existing particle at $\tau = \beta$ and erase that particle (the worm head moves backward in time) until reaching $\tau = 0$. The move is completed. The particle number is restored and a new configuration is sampled.

1. Sample configurations

Let us start from Fig. 2. After the bond update, a certain number bond operators is now activated.

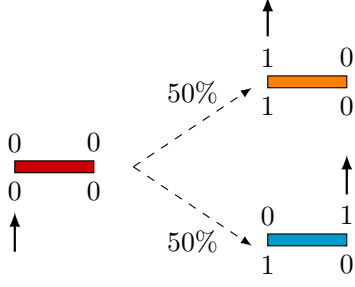


FIG. 3. Worm update part I. The upward arrow represents the head of the worm as it enters/leaves the site forward in time.

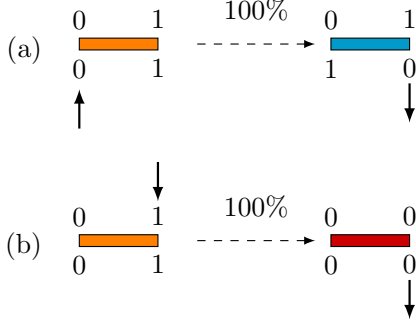


FIG. 4. Worm update part II. The upward arrow represents the head of the worm as it enters/leaves the site backward in time.

V. OBSERVABLES

Starting with the partition function

$$\begin{aligned} Z_{fi} &= \langle f | T(\beta) | i \rangle \\ &= \sum_{[b,s]} \langle f | e^{-\varepsilon H} | s_{N_b-1} \rangle \dots \langle s_2 | e^{-\varepsilon H} | s_1 \rangle \\ &\quad \times \langle s_1 | e^{-\varepsilon H} | i \rangle, \end{aligned} \quad (62)$$

Let us consider the transition amplitude between a state at time slice t and a state at time slice $t+1$:

$$\langle s_{t+1} | e^{-\varepsilon H} | s_t \rangle \simeq \langle s_{t+1} | \left(1 - \varepsilon \sum_b H_b \right) | s_t \rangle \quad (63)$$

If the particle does not move, $|s_{t+1}\rangle = |s_t\rangle$,

$$\langle s_{t+1} | e^{-\varepsilon H} | s_t \rangle \approx 1 + \mathcal{O}(\varepsilon^2) \quad (64)$$

If the particle hops to one of its nearest neighbor sites,

$$\langle s_{t+1} | e^{-\varepsilon H} | s_t \rangle \approx -\varepsilon \sum_b \langle s_{t+1} | H_b | s_t \rangle \quad (65)$$

every single hop (bond operator is activated) introduces a explicit factor of $\varepsilon = \beta/L_\tau$ into the configuration's

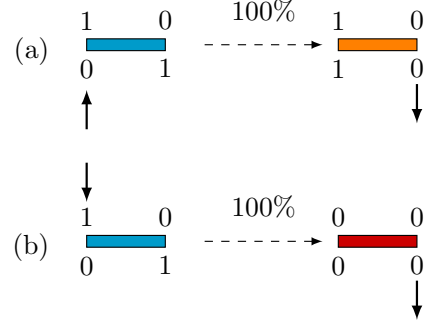


FIG. 5. Worm update part III. The upward arrow represents the head of the worm as it enters/leaves the site backward in time.

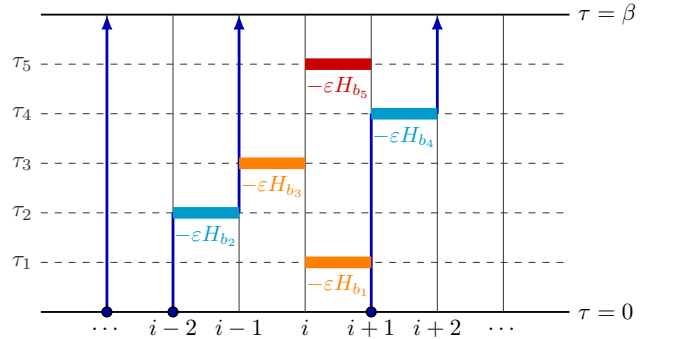


FIG. 6. Worm update part III. The upward arrow represents the head of the worm as it enters/leaves the site backward in time.

weight. If a full worldline configuration across the entire imaginary time axis contains exactly N_{op} bond operators, then

$$W([b, s]) \propto \varepsilon^{N_{\text{op}}} = \left(\frac{\beta}{L_\tau} \right)^{N_{\text{op}}} \quad (66)$$

We define the corresponding projected *thermal energy matrix* E in the same subspace, with matrix elements

$$\begin{aligned} E_{fi} &= \langle f | H e^{-\beta H} | i \rangle \\ &= -\frac{\partial}{\partial \beta} Z_{fi} = -\frac{\partial}{\partial \beta} \sum_{[b,s]} W([b, s]) \end{aligned} \quad (67)$$

Then we have

$$\begin{aligned} -\frac{\partial}{\partial \beta} \left(\frac{\beta}{L_\tau} \right)^{N_{\text{op}}} &= -N_{\text{op}} \left(\frac{\beta^{N_{\text{op}}-1}}{L_\tau^{N_{\text{op}}}} \right) \\ &= -\frac{N_{\text{op}}}{\beta} W([b, s]) \end{aligned} \quad (68)$$

It follows that

$$E_{fi} = \sum_{[b,s]} -\frac{N_{\text{op}}}{\beta} W([b, s]) = -\frac{\langle N_{\text{op}} \rangle}{\beta} \quad (69)$$

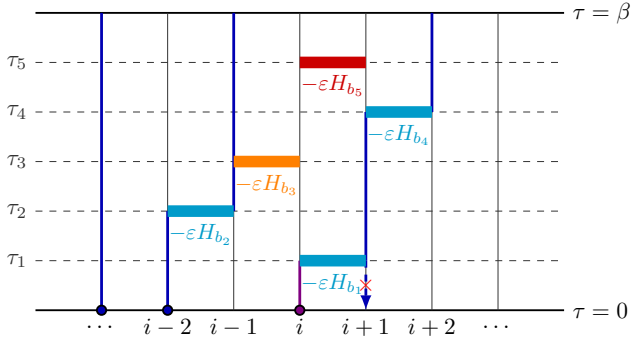


FIG. 7. Worm update part III. The upward arrow represents the head of the worm as it enters/leaves the site backward in time.

So MC sweep that yields a new configuration, we just need to count the number of bond operator!!!

VI. FERMIONIC SIGN PROBLEM

For detailed literature review, see Ref. [3]. Sign problem means the weights of configurations in the QMC simulation become negative or even complex, and therefore the configurational weights in the Monte Carlo process cannot be further interpreted as classical probabilities, and the minus or even complex sign will rendered the QMC simulation with exponential complexity for obtaining the data with the same quality compared with the simulations without sign problem.

The origin of the sign problem has no universal explanation, but it is believed to come from the quantum mechanic properties of the constituent particles in the many-body systems, given them fermions. While in fermionic systems, negative weights usually arise from the Pauli exclusion principle (Fig. 8). $|i\rangle = |f\rangle = (0, 1, 1, 0) \equiv c_1^\dagger c_2^\dagger |0\rangle$. There are four bond operator involved $H_{b_4} H_{b_3} H_{b_2} H_{b_1} \propto (c_2^\dagger c_0)(c_1^\dagger c_3)(c_0^\dagger c_1)(c_3^\dagger c_2)$. Note that for fermion, the anti-commutation relation reads $\{c_i, c_j^\dagger\} = \delta_{ij}$; $\{c_i, c_j\} = \{c_i^\dagger, c_j^\dagger\} = 0$.

$$\begin{aligned} \langle f | (c_2^\dagger c_0)(c_1^\dagger c_3)(c_0^\dagger c_1)(c_3^\dagger c_2) | i \rangle &= \langle f | c_0(c_1^\dagger c_3)(c_0^\dagger c_1)c_3^\dagger n_2 | i \rangle \\ &= \langle f | c_0 c_1^\dagger (c_0^\dagger c_1) (1 - n_3) n_2 | i \rangle \\ &= -\langle f | (1 - n_0) n_1 (1 - n_3) n_2 | i \rangle \\ &= -\langle f | (1 - 0) 1 (1 - 0) 1 | i \rangle = -1 \end{aligned}$$

Therefore, for a particular configuration, its weight $W([b, s])$ can be negative.

A. Strategy to deal with the sign problem

We treat our fermions as hardcore bosons (theoretical particles or systems of particles that behave like regular

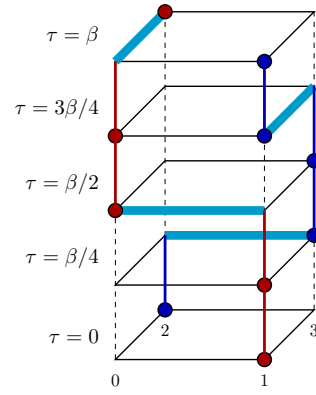


FIG. 8. An example of a configuration where particle exchange leads to negative weights. In 1D, Particle A cannot get past Particle B without jumping on top of it. Because the Pauli block prevents them from occupying the same site, they can never swap places. But in a higher dimension, they can go around each other. These particles are indistinguishable. But for demonstration, they have different colors.

bosons but repulse so strongly that no two particles can occupy the exact same quantum state).

The algorithm we discussed up to this point applies to the hardcore boson.

$$Z_{fi}^{HB} = \langle f | e^{-\beta H} | i \rangle = \sum_{[b, s]} |W([b, s])| \quad (70)$$

To obtain the physics of fermions, we have to push the actual sign of the configuration into the observable itself.

$$Z_{fi}^F = \sum_{[b, s]} \text{sign}([b, s]) |W([b, s])| = \langle \text{sign}[b, s] \rangle_{|W|} \quad (71)$$

Also

$$\begin{aligned} E_{fi}^F &= \sum_{[b, s]} \text{sign}([b, s]) \left(-\frac{N_{\text{op}}}{\beta} \right) |W([b, s])| \\ &= -\frac{1}{\beta} \langle \text{sign} \times N_{\text{op}} \rangle_{|W|} \end{aligned} \quad (72)$$

We first need to decide on creation ordering, for example $c_i^\dagger c_j^\dagger c_k^\dagger |0\rangle$ such that $i < j < k$

Although from the physics' perspective, these two particles are identical. However, the code will still track each particle independently.

At time $\tau = 0$ (the floor of your simulation), your code reads the $|i\rangle$ array. Let's say particles are at sites 1, 3, and 5. We agree on a strict convention: we always write the creation operators in ascending numerical order. $|i\rangle = c_1^\dagger c_3^\dagger c_5^\dagger |0\rangle$.

Let say, after the MC sweep, the particle from site 1 ends up at site 5, the one from 3 ends up at 1, and the one from 5 ends up at 3, i.e., $|f\rangle = c_5^\dagger c_1^\dagger c_3^\dagger |0\rangle$, which does not obey the strict ordering we set. We need 2 swaps to turn $5, 1, 3 \rightarrow 1, 3, 5$. Each swap would introduce -1

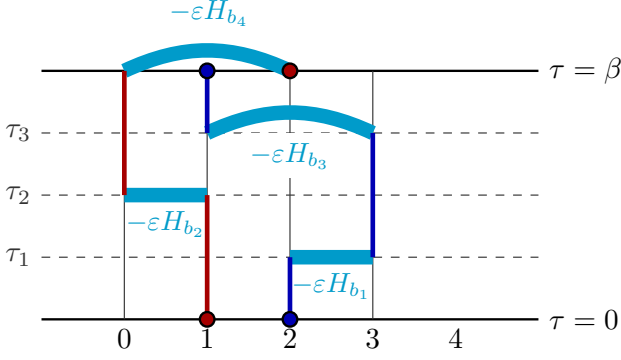


FIG. 9. Fermion swap between the two particles shown in Fig. 8

fermion sign. Based on the initial state, final state and the map of active bond operators, it is possible to track where each particle will land. Afterwards, we perform a counter to find the minimum number of swaps.

$$\text{sign}([b, s]) = (-1)^{\text{swap}}. \quad (73)$$

VII. TESTING

For a free particle on a 2D square periodic lattice, the kinetic energy is solved by moving to momentum space. The energy eigenvalues are given by the dispersion relation:

$$E = 2t [2 - \cos(k_x) - \cos(k_y)] - 2N_b t \quad (74)$$

Because your lattice is L with periodic boundaries, the allowed momenta are quantized into exactly: $k_x, k_y \in \{0, \dots, 2\pi n/L\}$ for $n \in \{0, 1, \dots, L-1\}$.

For simplicity, let assume $t = 1.0$.

L	N_{particle}	β	MC	Exact
4	2	1.0	-63.573(4)	-63.56914
4	3	1.0	-62.663(5)	-62.66398
4	4	1.0	-61.255(5)	-61.25074
4	5	1.0	-59.294(4)	-59.29942
4	6	1.0	-56.787(4)	-56.78704
4	7	1.0	-53.693(4)	-53.69263
4	8	1.0	-49.999(4)	-49.99806

TABLE I. Hardcore Boson, $t = +1$

L	N_{particle}	β	$\langle \text{sign} \rangle$	MC	Exact
4	2	1.0	0.12839	-61.869(7)	-61.864
4	3	1.0	0.03751	-59.857(27)	-59.861
4	4	1.0	0.00534	-57.302(214)	-57.107
4	5	1.0	0.00246	-55.137(458)	-55.593
4	8	1.0	0.00246	-44.841 $\pm$ 2.964	-43.488

TABLE II. Fermion

L	N_{particle}	β	$\langle \text{sign} \rangle$	MC	Exact
4	1	0.2	1	-64.002(4)	-64.0
4	2	0.2	0.59523	-61.201(5)	-61.210
4	3	0.2	0.44433	-58.894(6)	-58.898
4	4	0.2	0.12839	-54.911(12)	-54.914
4	5	0.2	0.18135	-51.232(17)	-51.261
4	6	0.2	0.14668	-47.654(19)	-47.676
4	7	0.2	0.12576	-43.799(16)	-43.794
4	8	0.2	0.12929	-40.240(17)	-40.236

TABLE III. Fermion, $\beta = 0.2$

L	N_{particle}	β	MC	Exact
4	1	1.0	-62.825(26)	-62.825
4	2	1.0	-60.956(19)	-60.981
4	3	1.0	-58.663(91)	-58.840
4	4	1.0	-57.202(159)	-57.308
4	5	1.0	-55.223(309)	-55.305

TABLE IV. Pi-flux fermion for $\beta = 1.0$

To test the diagonal bond operator update, for a single particles. Total number of bonds is $2L^2$. With only bond operator update, the particle cannot move, it permanently sits on 1 site. So only 4 bonds are single-occupied bonds and the rest $2L^2 - 4$ are empty bonds. The total weight is

$$(2L^2 - 4) \times 2t + 4 \times 1t \quad (75)$$

Then

$$\langle n_{\text{diag}} \rangle = 4\beta(L^2 - 1) \times t \quad (76)$$

VIII. ERROR ANALYSIS: BOOSTRAP METHOD

Because QMC samples configurations sequentially (Markov Chain), consecutive samples are correlated. Autocorrelation time measures the number of steps QMC required to generate a new, statistically independent configuration (the new configuration "forgets" its initial configuration).

Because samples are correlated, statistical analysis directly on the raw dataset will fail. Reblocking (or binning) is necessary to group QMC-produced data into larger, independent blocks to obtain accurate error estimates (see Fig. 10 for a sample of raw dataset before processing).

Then the "bootstrap" method can be used to estimate statistical uncertainties when observables are correlated, nonlinear, or derived from fitting procedures. For example, Suppose your QMC gives these 5 measurements: [10, 11, 9, 12, 8]. Of course it is straightforward to obtain mean = 10 and standard deviation = 1.41.

However, bootstrap asks: "If I repeated the whole QMC simulation again, how much could this average fluctuate?"

L	N_{particle}	β	MC	Exact
4	1	0.1	-62.318(5)	-62.326
4	2	0.1	-59.908(8)	-59.904
4	3	0.1	-57.145(9)	-57.144
4	4	0.1	-54.101(8)	-54.098
4	5	0.1	-50.701(8)	-50.695
4	6	0.1	-47.047(11)	-47.038
4	7	0.1	-43.232(12)	-43.235
4	8	0.1	-39.285(12)	-39.292

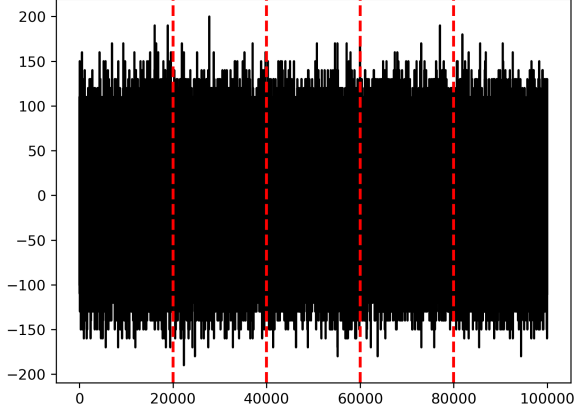
TABLE V. Pi-flux fermion for $\beta = 0.1$ 

FIG. 10. Sample raw dataset of a Monte Carlo simulation the first 100000 values

Since we do not have many independent simulations, bootstrap creates pseudo datasets by resampling the original data. For example, one replacement is $[10, 10, 8, 12, 9]$, then calculate the mean, $x_1^* = 9.80$. Resampling again, $[11, 12, 8, 12, 10]$, $x_2^* = 10.6$. Repeating this many times, we may get something like $[9, 80, 10.6, 10.2, 9.40, \dots]$. These numbers fluctuate and the fluctuation tells us the uncertainty. Now we need to be careful that bootstrap datasets are fake. It cannot create new physics. It only improves how accurately you estimate the uncertainty from the existing data.

IX. TWO-COMPONENT FERMION

Our lattice model is constructed in the Fock space formulation, using fermionic annihilation and creation operators $\psi_{\sigma,x}$ and $\psi_{\sigma,x}^\dagger$ where x is the lattice site on a periodic one-dimensional spatial lattice with L sites, and $\sigma = \uparrow, \downarrow$ are the two different fermion species. The lattice Hamiltonian can be arranged in two parts: a free Hamiltonian representing the kinetic part, H_0 , and an

interaction term, H_{int} , representing the onsite interaction

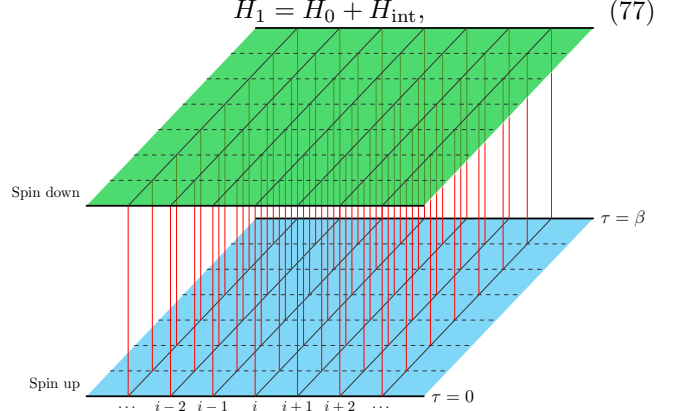


FIG. 11. Illustration of a continuous-time path-integral configuration of two-component fermion

where

$$H_0 = -t \sum_{\langle i,j \rangle, \sigma} (\psi_{\sigma,i}^\dagger \psi_{\sigma,j} + \psi_{\sigma,j}^\dagger \psi_{\sigma,i}) \quad (78)$$

$$- \psi_{\sigma,i}^\dagger \psi_{\sigma,i} - \psi_{\sigma,j}^\dagger \psi_{\sigma,j} + 2 \times \mathbb{1}) \quad (79)$$

$$H_{\text{int}} = U_s \sum_i (n_{\uparrow,i} n_{\downarrow,i} - \mathbb{1})$$

where the number operator $n_{\sigma,i} = \psi_{\sigma,i}^\dagger \psi_{\sigma,i}$. For now, we can consider $U_s > 0$ for repulsive interaction (e.g., Coulomb)

As we can see from Fig. 11, the interaction term acts as a diagonal bond connecting the two layers. We can quickly adapt the CTQMC to take this into account.

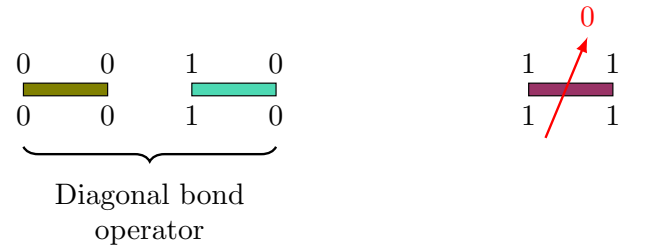


FIG. 12. Possible bond operators acting between layers, where the 0 and 1 represent the fermion occupation at a particular site.

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